

Bacterial Genus Classification from 16S rRNA Sequences

A reproducible machine learning pipeline for classifying bacterial genera using k-mer feature engineering, dimensionality reduction, and GPU-accelerated gradient boosting.

D A T A S E T

Source	Raw sequences	After validation
RDP Trainset 18	~21,000	18,815
16S rRNA gene sequences	~3,000 genera before filtering	2,380 records excluded

B E S T P E R F O R M I N G P I P E L I N E

- 01 **FASTA parsing** — Biopython · taxonomy validation · top-100 genera selection
- 02 **K-mer feature engineering** — CountVectorizer → sparse feature matrices
- 03 **Dimensionality reduction** — TruncatedSVD · 128 components
- 04 **Class balancing** — SMOTE(k_neighbors=1) · applied to training set only
- 05 **Classification** — XGBoost · tree_method="hist" · device="cuda" · NVIDIA RTX 5070

R E S U L T S · T O P - 1 0 0 G E N E R A

Accuracy	Macro F1	Weighted F1
98.5%+	~0.98	~0.99

I N S T A L L A T I O N

1. Create a virtual environment

```
python -m venv .venv
```

2. Activate

```
.venv\Scripts\activate
```

3. Install dependencies

```
pip install -r requirements.txt
```

TRAINING

Run the final pipeline

```
python scripts/train_final_xgb_svd_smote.py \  
  --data-path path/to/trainset18_062020.fa
```

Save artifacts

```
python scripts/train_final_xgb_svd_smote.py \  
  --data-path path/to/trainset18_062020.fa \  
  --save-artifacts
```

Output artifacts

```
vectorizer.joblib      svd.joblib      label_encoder.joblib  
xgboost_model.json    classification_report.txt  metrics.json
```

REPOSITORY STRUCTURE

```
bacterial-classification-16s-rna/  
├── notebooks/  
├── scripts/  
│   ├── analyze_classes.py  
│   ├── check_data.py  
│   ├── test_features.py  
│   ├── train_baseline.py  
│   ├── train_xgb_gpu.py  
│   └── train_final_xgb_svd_smote.py
```

```
├── src/bacterial_classifier/
│   ├── data.py
│   ├── features.py
│   ├── config.py
│   └── paths.py
├── outputs/models/
├── requirements.txt
└── README.md
```

TECHNOLOGIES

Python · Biopython · Scikit-Learn · XGBoost · Imbalanced-Learn · NumPy
· SciPy · TruncatedSVD · SMOTE · CUDA GPU Acceleration

FUTURE WORK

- FastAPI inference service
- Web-based sequence classification demo
- Docker deployment on DigitalOcean
- Model explainability and feature importance analysis
- Additional taxonomy levels — family, order, class

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Machine Learning · Bioinformatics · Data Science · GPU-Accelerated Classification